

Ferry Rachman Kurniawan

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[GitHub](#) | [Portfolio](#)

Profile

Bachelor of Computer Engineering graduate from Bina Nusantara University, specializing in Internet of Things (IoT) and embedded systems. Experienced in designing end-to-end IoT solutions, from hardware prototyping using ESP32, Arduino, sensors, and actuators to developing cloud backends and real-time communication using MQTT and REST APIs. Proven ability to integrate embedded devices, mobile applications, and cloud services into scalable and reliable systems, with hands-on experience in IoT projects. Strong problem solver with experience in device–cloud communication, system optimization, and multi-layer security design.

Education

Bina Nusantara University

2022 - 2026

Bachelor of Computer Engineering – GPA 3.38

Achievements:

- Finalist Gemastik 2025
- Top 50 APAC - Google Solution Challenge 2025
- Special Winner 2024 - Binus IoT Expo
- Best Microcontroller Final Project

Experience

Zegen Solusi Mandiri

Feb 2025 – Jul 2025

Mobile Programmer | Internship

- Developed and maintained 2+ cross-platform mobile applications for Android and iOS using Flutter.
- Implemented 10+ features and bug fixes across multiple sprints, improving app stability and performance.
- Collaborated with cross-functional team members to ensure on-time delivery of features and fixes.

Projects

LockGuard Pro Smart IoT Lock for Medicine & Vaccine Distribution | IoT & Mobile Developer

- Technologies used: ESP32, GNSS u-blox NEO-M8, OV2640 Camera, Flutter, Firebase, Cloud AI
- Developed a smart electronic lock system integrating 5+ hardware components (ESP32, GNSS, camera, solenoid lock, TFT LCD).
- Implemented 3-layer authentication system (PIN, QR code, face recognition) for secure access control.

- Designed geofencing-based unlocking system using GNSS real-time location validation.
- Built Flutter mobile application with real-time monitoring, access history logs, and notifications.
- Integrated cloud backend enabling real-time data synchronization and remote control.

Smart Medicine Dispenser | Full Stack Developer

[🔗 GitHub Link Mobile](#) | [🔗 GitHub Link ESP](#)

- Technologies used: Flutter, Golang, Azure, ESP32, Arduino, FCM, RESTful API, MQTT.
- Built an end-to-end IoT medicine dispenser system integrating mobile app, cloud server, and ESP32 hardware.
- Implemented MQTT communication for real-time device-to-cloud data exchange.
- Developed automated dosage dispensing system based on scheduling and user input.
- Designed mobile application with 5+ core features including scheduling, monitoring, notifications, and alerts.
- Integrated FCM notifications for medicine reminders and emergency alerts.

Organization Experience

Bina Nusantara English Club (BNEC)

Manager of Information Technology | BNEC Presidency **2024**

- Maintained and updated BNEC's official website and digital platforms.
- Designed and delivered web development training for internal staff.
- Provided technical support for organizational activities and events.

Vice Project Operational | New Member Recruitment **2023**

- Led 2 divisions in planning and executing large-scale recruitment events.
- Coordinated operations for multiple activities including Counseling Class, UKM Expo, Comprehensive Presentation, and Welcoming Party.
- Implemented a seamless digital registration process for new members.

Technical & Soft Skills

Framework: Flutter, Laravel, Arduino, Qt

Languages: Dart, PHP, C/C++

Hardware & Embedded: ATmega328P, Arduino UNO, ESP32

3D Design: Fusion 360

Tools & Other: Git, REST API, Firebase, Bloc, Getx

Soft Skills: Team work, Adaptability, Time management, Analytical thinking, Leadership